1) Project Title:

Real - Time Speech Emotion Recognition (SER) system using deep learning and data augmentation

2) Student Information:

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3) Project Overview:

The project aims to develop a real time speech emotion recognition system (SER) using deep learning and data augmentation tools. Currently, machine - human interfaces are capable of communicating and coordinating various tasks however there is a clear limitation that hinders the natural empathetic communication that a human would have with another human. The SER project attempts to bridge this gap so that digital tools become more empathetic and emotionally responsive. This would allow for more accurate and productive communication in various applications such as customer service automation, digital mental health monitoring and education/E- learning services.

4) Background and Motivation:

There has been rapid growth in LLMs in the current decade, however, there seems to be a clear gap in the emotional discourse between humans and machine communication. Human communication relies heavily on prosodic features, such as pitch changes, speech rate, and spectral energy which carry key context beyond just literal words. Traditional speech processing algorithms like Support Vector Machines (SVMs) or Hidden Markov Models (HMMs) were effective in quiet lab conditions, however these classical systems drop in performance when there is background noise in presence. Modern Deep learning architectures offer significantly stronger feature generalisation, but many heavy models (e.g., deep 2D CNNs or Transformers) introduce latency constraints that prevent low-cost deployment on real-time streaming audio. This project was selected to solve the balance between speed and accuracy. We will utilise a lightweight MFCC (Mel-Frequency Cepstral Coefficient) audio feature with simple Multi - Layer Perceptron (MLP) in PyTorch, hence, the system can process speech much more quickly in real time. Additionally, adding noise augmentation during training ensures the model will remain reliable when capturing live audio from a microphone in everyday environments.

5) Proposed Methodology:

This project uses Python alongside Librosa and PyTorch to build a speech emotion recognition system. The dataset used is a subset of RAVDESS database, selecting key audio samples to train the model effectively. Using Librosa, raw .wav audio files are converted into time-series signals and paired with emotional labels like happy, sad, or angry.

To make the system reliable in real-world environments, several data augmentation techniques are applied to the audio. Synthetic background noise, such as Gaussian and white noise, is added directly to the signals to help the model handle room static. Additionally, spectrogram shifting and masking techniques adjust pitch and time frames, preventing the system from simply memorising specific voices. Finally, these processed audio features are fed into a Multi-Layer Perceptron (MLP) neural network, creating a robust classifier capable of accurately detecting emotions even when background noise is present.

6) Expected Outcomes:

The project will deliver a working Python program that listens to live audio through a microphone and updates an emoji on the screen (such as happy, sad, or angry) to match the current emotion in your voice in real time. Below the emoji, a live percentage counter will display how confident the system is in its prediction for each emotion. The overall performance will be measured using standard machine learning metrics like test accuracy and confusion matrices to compare quiet audio against noisy recordings. Finally, all the Python code, trained PyTorch model files, and setup instructions will be published in a public GitHub repository.

7) Timeline:

Week	Task
1-2	Real Time Speech Emotion Recognition (SER) Project chosen
3-5	Filtering the **RAVDESS** dataset, setting up **PyTorch**, implementing **librosa** loading, setting up the **Data Augmentation Tools** (**Noise Addition** and **Spectrogram Shifting**) and extracting **MFCC** features
6-9	**MLP** Architecture & Initial Implementation: Build the PyTorch MLP (Multi-Layer Perceptron) **neural network**, train it on the extracted MFCC features, and connect it to the real-time audio GUI with live emoji updates.
10-11	**MLP Optimisation & Evaluation:** Fine-tune the MLP model's hyperparameters (layers, learning rate, loss functions), evaluate classification performance on clean vs. noisy audio, and refine the real-time confidence analyser.
12-13	**Documentation & Final Delivery:** Complete the final project report, finalise public GitHub repository with trained MLP model weights, and prepare the project demonstration.

8) References:

1) Barhoumi, C., & BenAyed, Y. (2025). Real-time speech emotion recognition using deep learning and data augmentation: Deep learning-based real-time speech emotion recognition: C. Barhoumi, Y. B. Ayed. *Artificial Intelligence Review*, *58*(2), 1–41. https://doi.org/10.1007/s10462-024-11065-x

2) Hama Saeed, M. (2023). Improved Speech Emotion Classification Using Deep Neural Network. *Circuits, Systems & Signal Processing*, *42*(12), 7357–7376. https://doi.org/10.1007/s00034-023-02446-8

3) Alluhaidan, A. S., Saidani, O., Jahangir, R., Nauman, M. A., & Neffati, O. S. (2023). Speech Emotion Recognition Through Hybrid Features and Convolutional Neural Network. *Applied Sciences (2076-3417)*, *13*(8), 4750. https://doi.org/10.3390/app13084750